

Curriculum

Machine Learning & Artificial Intelligence

Module	Topics Covered
ML/AI Fundamentals	Types of ML (supervised, unsupervised, reinforcement), AI vs ML vs DL, use cases
Supervised Learning	Linear & logistic regression, decision trees, k-NN, SVM with Scikit-learn
Unsupervised Learning	K-Means clustering, PCA, dimensionality reduction, applications
Model Evaluation & Tuning	Train/test split, cross-validation, confusion matrix, precision/recall, hyperparameter tuning
Deep Learning Basics	Neural networks, activation functions, backpropagation, intro to TensorFlow/Keras
Natural Language Processing (NLP)	Text preprocessing, TF-IDF, sentiment analysis, text classification
Computer Vision Intro	Image processing, CNNs, object detection basics with pre-trained models
ML Model Deployment	Flask/FastAPI REST API, deploying model, real-world project demo